

UVM Verification

Architecture, verification logic, coverage, and findings

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Project journey

Specification to evidence

01

Understand

Read the BMU specification and record unclear behavior

02

Scope

Define the 21 legal modes and verification boundaries

03

Plan

Map requirements to scenarios, checks, and coverage

04

Build

Implement the UVM environment and independent model

05

Verify

Run directed tests, regression, and coverage analysis

06

Conclude

Separate stimulus closure from unresolved DUT behavior

Verification objective and scope

The verification question

Does the BMU produce the correct result and error for every approved request, at the correct cycle?

21 legal modes

Black-box

scan_mode = 0

- 18 BMU operations and 3 CSR data paths
- One-cycle registered result
- Result hold when valid_in = 0
- Invalid-control and recovery behavior
- Legal NOP and reset behavior

Rules that shaped the testbench

Latency

A request sampled in cycle N produces result_ff in cycle N+1

Hold

valid_in = 0 preserves the previous registered result

Error

Error depends on control legality, not on valid_in

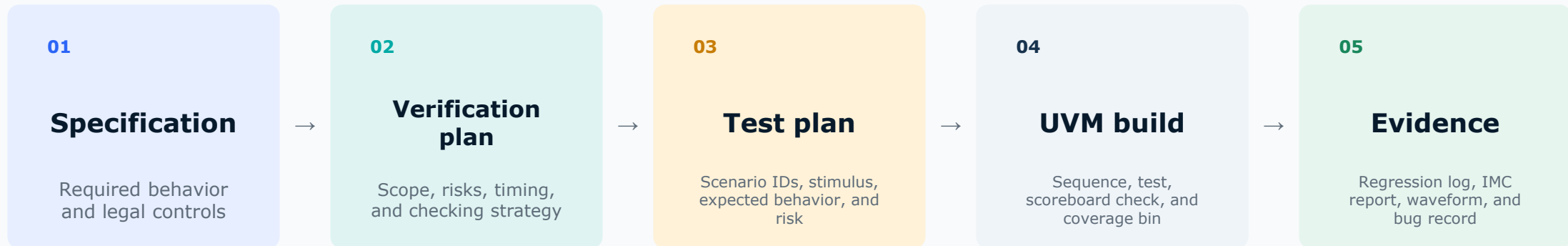
Reset

Synchronous rst_l = 0 clears result_ff and error

Corrections

CTZ(0) = 32 and invalid guards use OR logic

Planning and traceability



Traceability rule

Every important requirement must end in a reproducible check and visible evidence.

Coverage shows that the planned situation occurred. The scoreboard shows whether the DUT handled it correctly.

Scenario design process

01

Legal recipe

Start from $ap = 0$ and enable only the required fields

02

Behavior

Identify the result, timing rule, and ignored inputs

03

Risk

Choose boundaries that can change the implementation path

04

Stimulus

Use directed vectors, then limited random variation

05

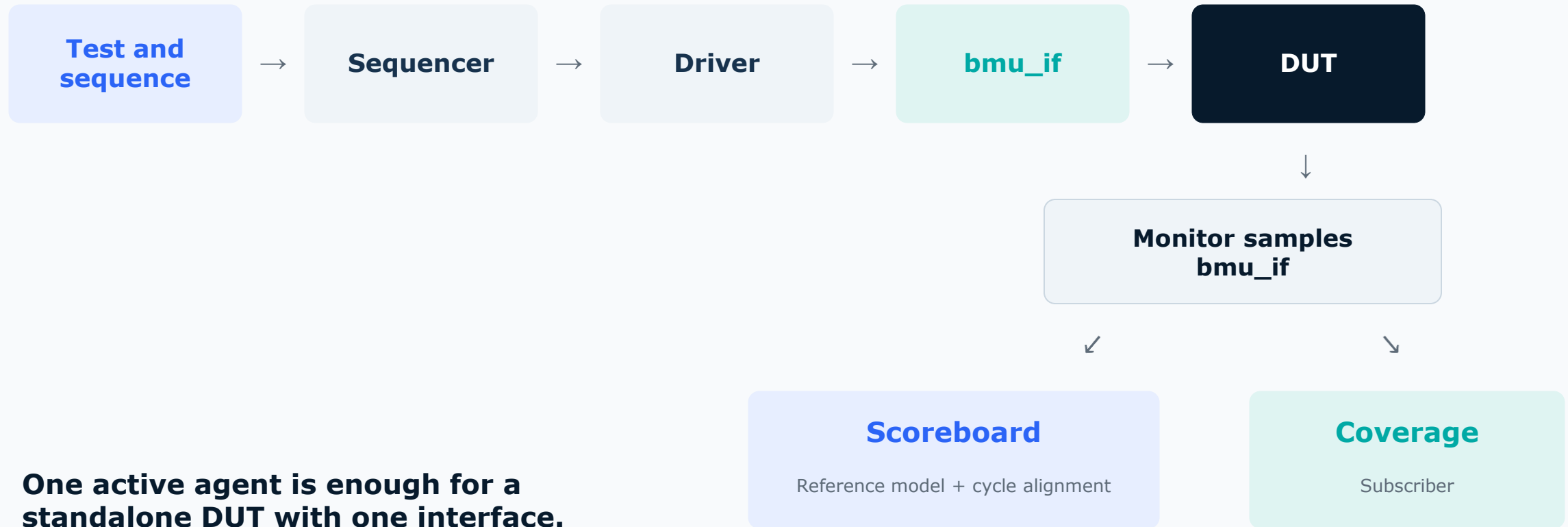
Evidence

Map the scenario to scoreboard checks and coverage bins

Design rule

Each scenario must verify a distinct requirement or risk. Repetition without new behavior was removed or grouped.

UVM architecture



One active agent is enough for a standalone DUT with one interface.

Component responsibilities

01 Sequence item

Carries BMU inputs and the outputs sampled by the monitor

02 Sequence and sequencer

Create legal, corner, mixed, control, and negative requests

03 Driver

Applies inputs through the driver clocking block

04 Monitor

Samples one black-box transaction and broadcasts it

05 Reference model

Computes expected behavior independently from the RTL

06 Scoreboard

Aligns the one-cycle result and compares current error

07 Coverage subscriber

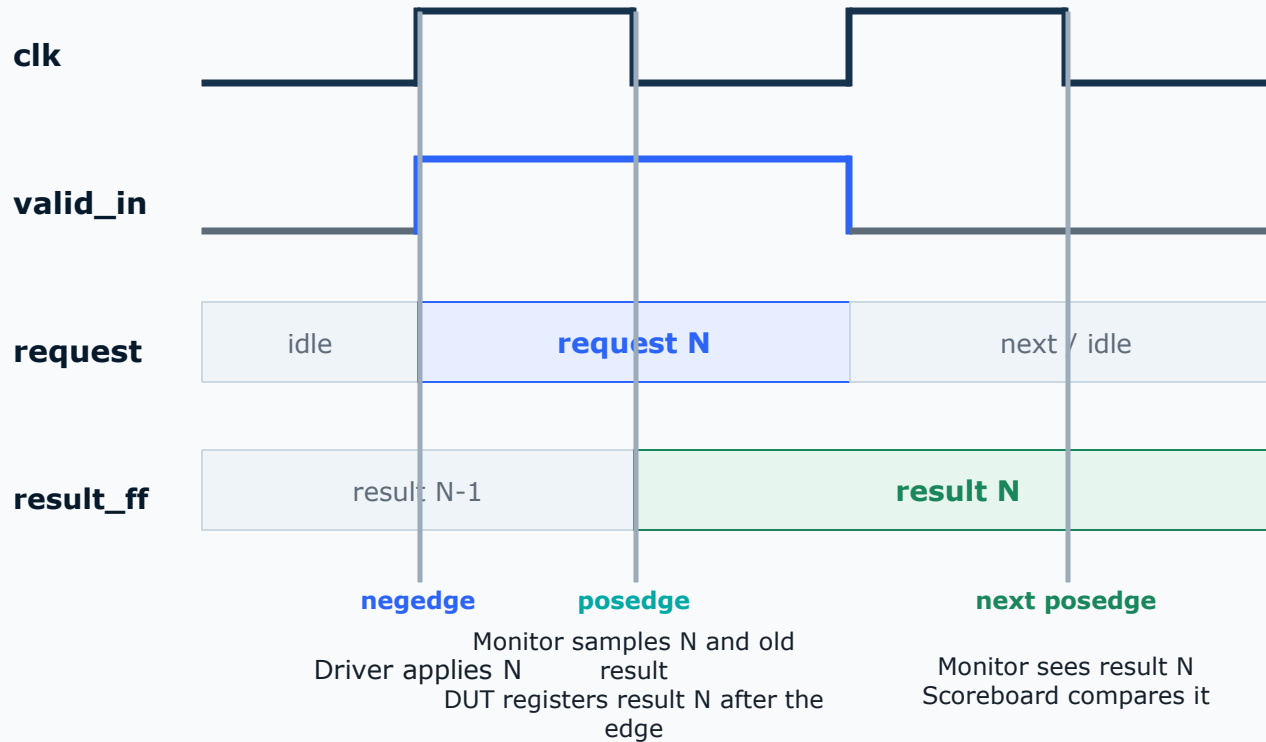
Classifies legal operations, operands, and error intent

08 Tests and regression

Select sequences, control objections, and collect evidence

Clocking and one-cycle latency

Timing model used by the checker



Sampling contract

Driver	Applies stable inputs at the negative edge
DUT	Accepts a valid request at the following positive edge
Monitor	Samples through cb_mon with input #1step
Scoreboard	Compares result N at the next monitor sample

Key point

The monitor samples before the posedge update, so the scoreboard checks request N when result N becomes observable at the next sample.

Scoreboard decision flow

1

Compare pending result

Uses the request saved from the preceding cycle

2

Decode current controls

Recognizes one exact legal recipe or a conflict

3

Check current error

Independent of valid_in

4

Save next expectation

Update, NOP zero, reset zero, or hold

Expected result state

valid_in = 1

Store the model result for the next cycle

valid_in = 0

Keep the previous expected result

rst_l = 0

Expect result_ff = 0 next cycle and error = 0 now

Stimulus strategy

01

Directed cases

Guarantee boundaries and corrected specification behavior

02

Limited random

Adds operand diversity without replacing required corners

03

Paired checks

Change only an ignored input to prove independence

04

Negative tests

Isolate one illegal-control reason at a time

Regression order

Reset / logic / shift and bit / arithmetic and compare / count and data / CSR / valid_in / errors

Corner cases with a clear purpose

Feature	Stimulus classes	Why it matters
Shift / rotate	0, 1–15, 16, 17–30, 31	Distinct amount behavior
BINV	0, 1–15, 16–30, 31 + original bit	Boundary and bit-toggle direction
CTZ	Exact expected counts 0–32	Every result identifies a different first set bit
CPOP	Seven count classes + four byte regions	Count range and full 32-bit contribution
SEXT.B	00, 7F, 80, FF + interior signs	Sign boundary and ignored upper bits
PACK / CSR	Asymmetric source values	Makes source-order reversal observable

Error verification

Legal request

ap = 0

+ required fields only

+ csr_ren_in = 0
unless CSR read is
selected

Expected error = 0

Five documented error intents

- CSR read with an AP operation
- SH2ADD without zba
- SUB with zba
- More than one primary operation
- Legal recipe plus one forbidden AP field

All five intents are exercised. One representative conflict is repeated with valid_in = 0.

Coverage model logic

Exact bins

Used when every value represents a distinct architectural result

21 legal modes

CTZ counts 0 through 32

Grouped bins

Used when several input values exercise the same functional risk

Shift	0 / low / 16 / high / 31
BINV	endpoints + lower and upper interiors
CPOP	count ranges + four source-byte regions
PACK / GREV	semantic operand and mode classes

Crosses

Kept only for meaningful interactions. Reset, hold, latency, and result correctness remain scoreboard checks.

Functional coverage closure

182 / 182

reported closure bins hit

Instruction selection

21 bins

Operation 21/21

Logic, shift, and bit

51 bins

Logic 16/16

Shift 15/15

SRA sign 2/2

BINV 6/6

Shift data 12/12

Arithmetic and count

76 bins

Compare 24/24

SUB 5/5

SH2ADD 3/3

CTZ 33/33

CPOP count 7/7

CPOP region 4/4

Data movement and control

34 bins

SEXT.B 10/10

PACK 3/3

GREV 2/2

CSR source 3/3

CSR data 9/9

Error reason 5/5

Error valid 2/2

Closure interpretation

All planned stimulus categories were observed. Functional coverage does not judge the DUT result and does not override scoreboard mismatches.

Authority

Coverage: stimulus closure
Scoreboard: correctness

Reported regression snapshot

XCELIUM CODE METRICS

100%

Instrumented block points
14/14

100%

Instrumented expression points
3/3

58.67%

Toggle 1045/1781

SCOREBOARD

845

monitored samples

1,689

total checks

222

failed checks

Check outcome

1,467 pass

222 fail

205 result mismatches

17 error mismatches

Recorded BMU_SB_SUMMARY

observed=845 total_checks=1689 pass=1467 fail=222

result_fail=205 error_fail=17 UVM_FATAL=0

Regression findings documented in the test plan

CTZ

6 / 36 valid checks passed

Incorrect trailing-zero results across many counts

PACK

2 / 49 checks passed

A and B halfword lanes are reversed

CPOP

15 / 56 checks passed

Upper-half bits are not counted correctly

GREV mode 24

0 / 7 checks passed

Observed behavior matches a halfword swap, not REV8

MAX

4 / 26 checks passed

Equality passes; non-equal cases select the smaller signed operand

CSR write

0 / 13 checks passed

csr_imm selects the opposite source

Error path

NEG-05 exposed a missed GREV extra-field conflict. NEG-07 showed error incorrectly dropping when valid_in = 0.

Bug evidence: PACK source order

REPRODUCIBLE CASE

Operation	PACK
A	0x12345678
B	0xABCDEF01
Expected	{B[15:0], A[15:0]} 0xEF015678
Observed	0x5678EF01

Finding

The DUT concatenated the two halfwords in the opposite order.

Cycle-aligned mismatch

The request and response remain separated by the registered output latency

Cycle	Signal	Value / meaning
Sample N	valid_in = 1	PACK request accepted
Sample N	A / B	12345678 / ABCDEF01
Sample N+1	Expected	EF015678
Sample N+1	Observed	5678EF01
Sample N+1	error	0

The observed value equals {A[15:0], B[15:0]}, confirming reversed source order.

What the project proves

Closed

The planned 19-group coverage summary reached 100%. The consolidated regression checked timing, current-cycle error, reset, and hold behavior; the focused smoke test checked legal NOP.

Still open

DUT correctness, 736 toggle points, and the two CSR legality questions marked Blocked Q4 in the test plan.

Final evidence

- | | | |
|----|-------------------|---|
| 01 | Regression | BMU_SB_SUMMARY and the final pass/fail status |
| 02 | Functional | The reported 19 groups and 182 closure bins |
| 03 | Code | Block, expression, and toggle reports from IMC |
| 04 | Case study | Cycle-aligned PACK request, expected value, and observed mismatch |

Coverage closed. DUT sign-off remains open.